Result 1 (pair chain commutativity). [Topology foundations, singular chain functor on pair inclusions]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let X be a topological space. Let Q be a certified inclusion between pairs in X. Then

the chain square of Q commutes.

Result 2 (relative quotient maps commute). [Topology foundations, cokernel projection naturality]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let X be a topological space. Let Q be a certified inclusion between pairs in X. Assume the chain square of Q commutes (`h`). Then

$$C_*(Q) \circ \pi_{Q.\mathrm{source}} = \pi_{Q.\mathrm{target}} \circ C_*(Q.ambient).$$

Adapted from the cited source.

Result 3 (relative homology image membership). [Topology foundations, module map range membership]
Let K be a type. Let *field* be a field structure on K. Let H be a choice of coefficients and homology degree over K. Let X be a topological space. Let Q be a certified inclusion between pairs in X. Assume the chain square of Q commutes (`h`). Let x be an element of $H(Q.\mathrm{target})$. Let $f = H(Q)$. Let $U = \mathrm{range}(H(Q))$. Then

$$x \text{ belongs to } U \text{ if and only if } (\text{there exists } y : H(Q.\mathrm{source}),\ f(y) = x)\,.$$

Adapted from the cited source.

LLM-assisted Clean specimens and proofs (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.